

MASTITISGUARD AI — EXECUTIVE SUMMARY & PROJECT NARRATIVE

The Journey: Market Problem → Deep Understanding → Built Solution → Verified Design → Deployable at Scale

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Problem Statement Number: 26109

How to Read These Seven Documents

This project is presented as **seven linked documents**, each answering one specific question a reviewer will ask. Read in order, they trace a single journey: we found a real, sized problem in the market; we understood it well enough to know why existing solutions don't solve it for India; we built a working system around that understanding; we verified every design number rather than asserting it; and the result is **priced and architected to actually deploy at national scale**. Every figure that appears in one document — **cost, accuracy, timeline** — appears identically in every other document that references it.

Document	Journey Stage	Question It Answers
1. Executive Summary & Narrative (this document)	Overview	What is the whole story, in one read?
2. Market Research, Gap Analysis & Scaling Plan	1 — Problem from the market	Is this a real, sized problem, and why do existing solutions fail India?
3. Technical Dossier	2 — Understood completely	What is the clinical, economic, and engineering reasoning behind the design?
4. Bill of Materials	3 — Built solution	Exactly what hardware exists, at what cost, and to what build status?
5. Decision-Making Analysis & Proof of Concept	3 — Built solution	Why this approach over the alternatives, and what is actually working today?
6. ML Model Documentation	3 — Built solution	How does the predictive engine work, end to end?
7. Validation & Justification	4 — Verified with the market	Why this exact number, and not another one, for every design choice?

1. The Problem, From the Market

Bovine mastitis costs Indian dairying an estimated ₹7,000–₹11,500 crore a year. Subclinical mastitis (SCM) — the invisible stage, with no swelling and no visible sign in the milk — drives 58–70% of that loss and is present in roughly 45% of Indian dairy cattle at any given time, 15–40 times more common than the clinical disease a farmer can actually see. By the time a smallholder notices anything, the animal has already lost 3–4 kg/day of milk, a 3–7 day antibiotic withdrawal window is triggered, and treatment costs ₹3,500–₹6,000 — a significant sum against a 2–5 cow household's income. Global precision-dairy platforms (Allflex, Afimilk, CowManager, smaXtec) have proven that wearable, AI-based early warning works — but at ₹20,000–₹40,000 per cow, requiring mechanized in-line milking parlors and continuous connectivity, they are built for a herd size and infrastructure 95% of Indian dairy farmers do not have.

2. Understanding the Problem Completely

Before building anything, the team mapped the disease's full progression (subclinical drift → clinical flare-up → irreversible alveolar damage → blind quarters), the physiological signals that shift ahead of visible symptoms (skin temperature, rumination and activity, milk pH and yield), and the structural reason Western PLF systems don't transplant into India: they assume a fixed milking parlor and continuous internet, and neither exists at the 2–5 cow smallholder level. This understanding — detailed in the Technical Dossier — is what shaped every hardware and architecture decision that followed, and it is the same understanding that produced the competitive gap analysis and market sizing in the Market Research document.

3. The Solution Built

MastitisGuard AI is a three-unit, fully offline system:

- **Unit 1 — Wearable Collar (₹2,510/cow):** continuous skin temperature, activity and rumination proxy, and pasture mobility, worn by the animal.
- **Unit 2 — Shared Milking Station (₹2,110/shed, prototype grade):** per-session milk temperature, yield, and pH, tied to a scanned Cow ID.
- **Unit 3 — Shed Edge-AI Gateway (₹950/shed):** aggregates both units over a zero-connectivity ESP-NOW mesh and runs the risk-scoring engine locally.

The engine itself has two layers, both real and both running today: a deterministic per-cow rolling-baseline layer (each animal is compared only to her own 7–14 day history, never to a herd average), and a CatBoost classifier trained on top of that baseline data to produce a calibrated risk tier. The baseline layer needs no training data and works from day one of deployment; the CatBoost layer, trained and evaluated on physiologically-grounded synthetic data (ROC-AUC 0.964), sharpens that signal into SAFE / MEDIUM / CRITICAL tiers a farmer can act on. Both layers are documented in full in the ML Model Documentation and the Decision-Making document.

4. Verified, Not Just Asserted

Every number in this project — the dataset size, the class weights, the decision threshold, the fusion weight between sensor data and CMT confirmation — is traced back to a specific reason in the Validation & Justification document, not presented as an unexplained result. Equally, the Decision-Making document states plainly what is built and bench-verified today (sensing, ESP-NOW mesh, baseline computation, CatBoost training and offline export, the mobile app) versus what remains designed but not yet physically integrated (the pH probe, the CMT vision module) or not yet validated on real animals (field accuracy of the CatBoost tier, beyond its synthetic-data result). Nothing in this project claims more than has actually been demonstrated: the 1–5 day pre-clinical drift window is what has been shown on synthetic data today; the 7–14 day figure quoted in veterinary literature is the target Phase 1 real-farm validation is designed to reach, not a result already in hand.

5. Ready for Market Deployment at Scale

The entire architecture exists to make one number possible: ₹2,510 per additional cow, with the milking station and gateway shared across the whole herd — roughly an order of magnitude below the global entry price of ₹20,000–₹40,000/cow, and reachable through India's existing dairy cooperative network of 1.72 crore affiliated farmers rather than a dealer-led sales model. The three-phase scaling plan in the Market Research document takes this from a 200–500 cow pilot (Year 1, replacing synthetic training data with real

vet-confirmed outcomes) to a 50,000+ cow regional deployment (Year 3) integrated with national cooperative infrastructure.

6. The One-Line Summary

MastitisGuard AI takes a ₹7,000+ crore problem that Western precision-dairy platforms cannot solve at Indian smallholder economics, builds a working three-unit offline system around a deterministic per-cow baseline and a synthetic-data-trained CatBoost classifier, defends every design number against the alternative it beat, and prices the result at roughly a tenth of the global entry cost — ready to validate on real herds through India's cooperative dairy network.